Nova Analytics - EDA Report

Dataset: Real estate.csv

414 rows x 8 columns

Dataset Information

Shape

Rows	**414**
Columns	**8**

Data Types

Column	Data Type	Non-Null	Null	Unique
No	int64	414	0	414
X1 transaction date	float64	414	0	12
X2 house age	float64	414	0	236
X3 distance to the nearest MRT station	float64	414	0	259
X4 number of convenience stores	int64	414	0	11
X5 latitude	float64	414	0	234
X6 longitude	float64	414	0	232
Y house price of unit area	float64	414	0	270

Descriptive Statistics

Column	Mean	Median	Std Dev	Min	25%	50%	75%	Max
No	207.50	207.50	119.66	1.00	104.25	207.5	310.75	414.00
X1 transaction date	2013.15	2013.17	0.28	2012.67	2012.92	2013.17	2013.42	2013.58

X2 house age	17.71	16.10	11.39	0.00	9.02	16.1	28.15	43.80
X3 distance to the nearest MRT station	1083.89	492.23	1262.11	23.38	289.32	492.23	1454.28	6488.02
X4 number of convenience stores	4.09	4.00	2.95	0.00	1.0	4.0	6.0	10.00
X5 latitude	24.97	24.97	0.01	24.93	24.96	24.97	24.98	25.01
X6 longitude	121.53	121.54	0.02	121.47	121.53	121.54	121.54	121.57
Y house price of unit area	37.98	38.45	13.61	7.60	27.7	38.45	46.6	117.50

Missing Value Analysis

Total Missing Values: **0**

Column	Missing Count	Percentage
No	0	0.0%
X1 transaction date	0	0.0%
X2 house age	0	0.0%
X3 distance to the nearest MRT station	0	0.0%
X4 number of convenience stores	0	0.0%
X5 latitude	0	0.0%
X6 longitude	0	0.0%
Y house price of unit area	0	0.0%

No missing values detected in this dataset.

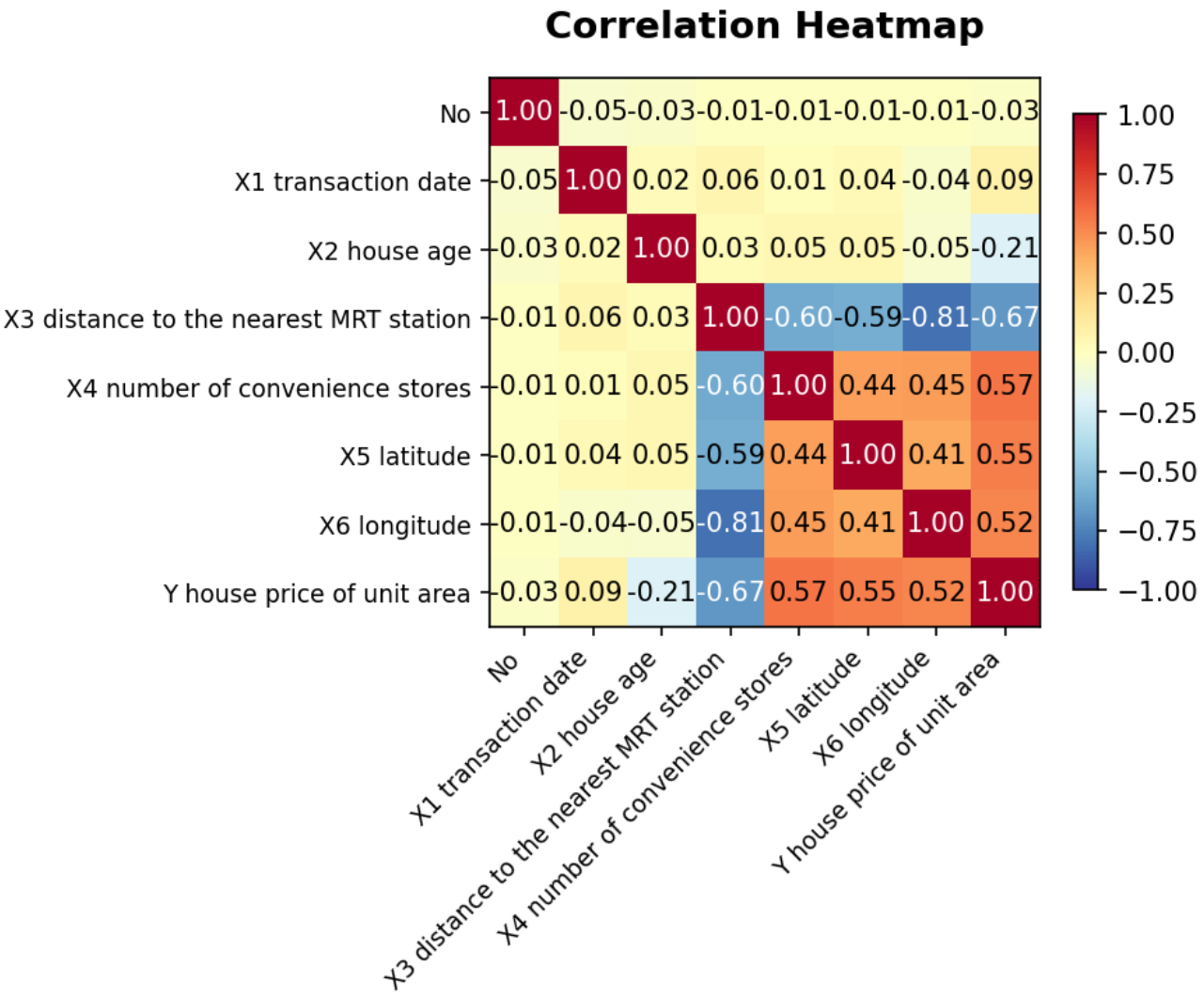
Duplicate Analysis

Duplicate Rows	0 (0.0%)
Total Rows	414

Correlation Analysis

Correlation Matrix

	No	X1 transacti..	X2 house age	X3 distance ..	X4 number of..	X5 latitude	X6 longitude	Y house pric..
No	1.00	-0.05	-0.03	-0.01	-0.01	-0.01	-0.01	-0.03
X1 transacti..	-0.05	1.00	0.02	0.06	0.01	0.04	-0.04	0.09
X2 house age	-0.03	0.02	1.00	0.03	0.05	0.05	-0.05	-0.21
X3 distance ..	-0.01	0.06	0.03	1.00	-0.60	-0.59	-0.81	-0.67
X4 number of..	0.01	0.01	0.05	-0.60	1.00	0.44	0.45	0.57
X5 latitude	-0.01	0.04	0.05	-0.59	0.44	1.00	0.41	0.55
X6 longitude	-0.01	-0.04	-0.05	-0.81	0.45	0.41	1.00	0.52
Y house pric..	-0.03	0.09	-0.21	-0.67	0.57	0.55	0.52	1.00

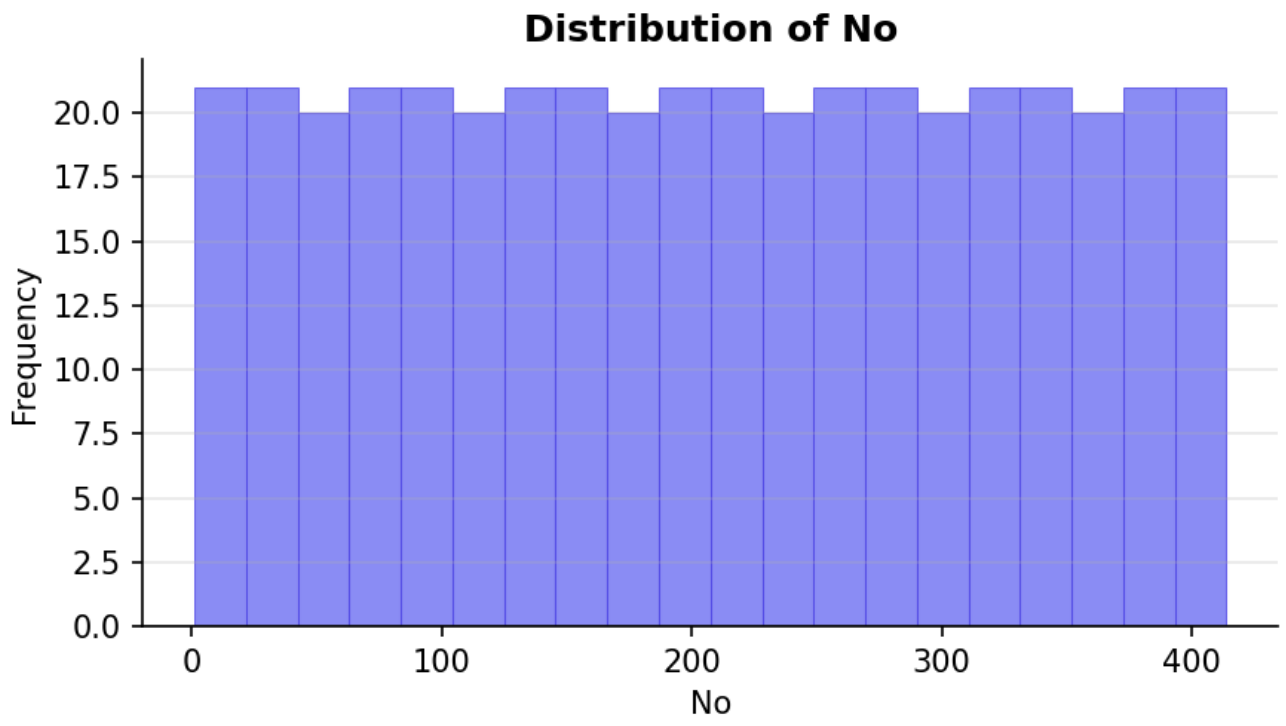


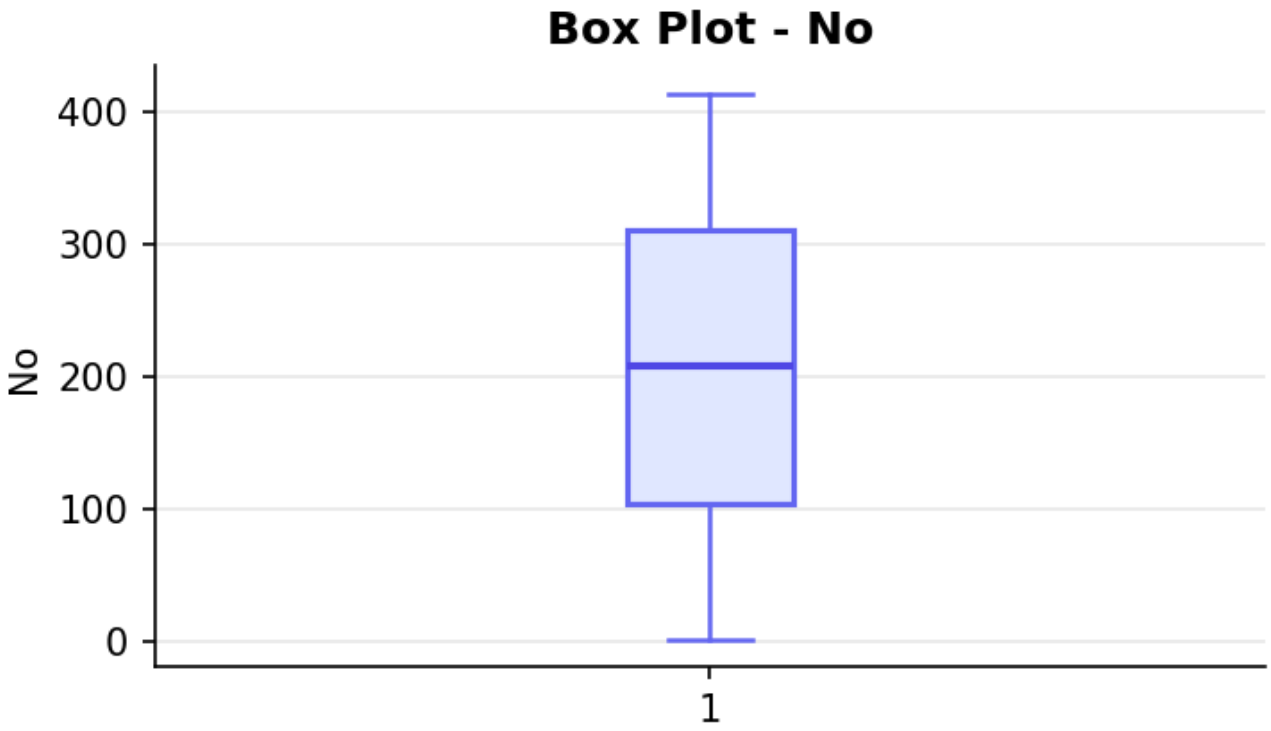
Outlier Analysis

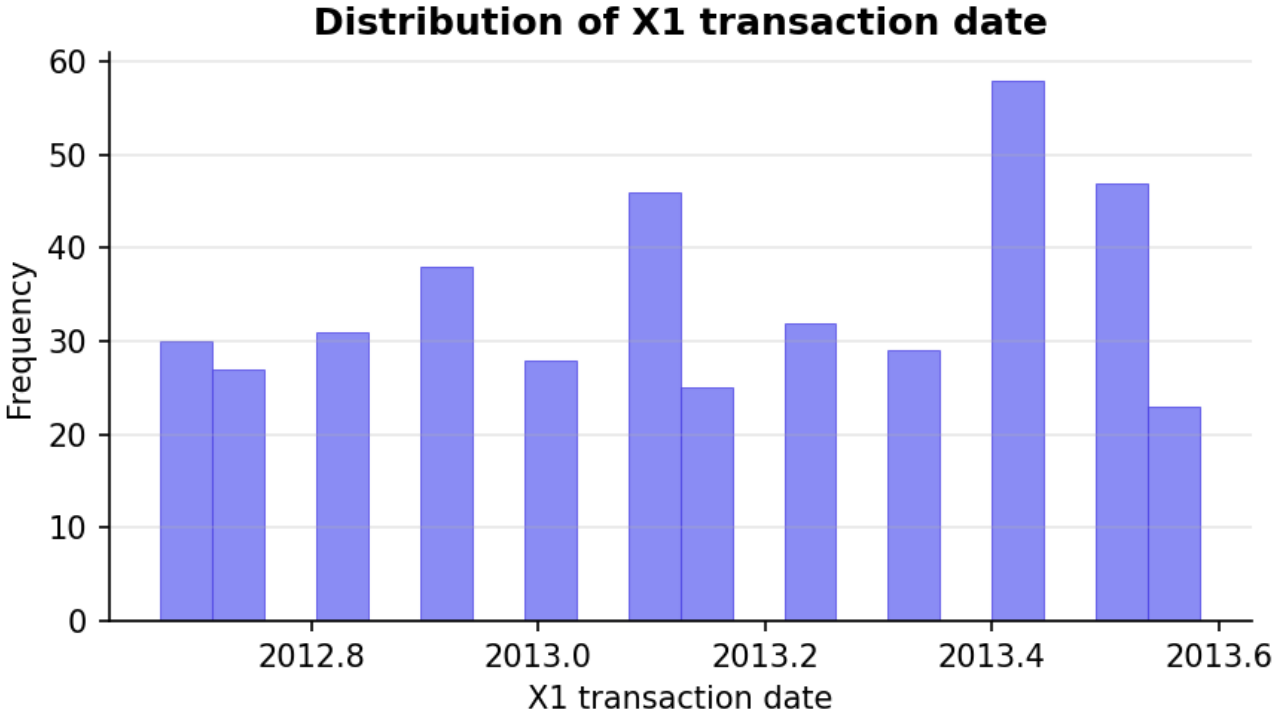
Column	Outliers	%	Q1	Q3	IQR	Lower Bound	Upper Bound
No	0	0.0%	104.25	310.75	206.50	-205.50	620.50
X1 transaction date	0	0.0%	2012.92	2013.42	0.50	2012.17	2014.17
X2 house age	0	0.0%	9.03	28.15	19.12	-19.66	56.84
X3 distance to the nearest MRT station	37	8.9%	289.32	1454.28	1164.95	-1458.11	3201.71

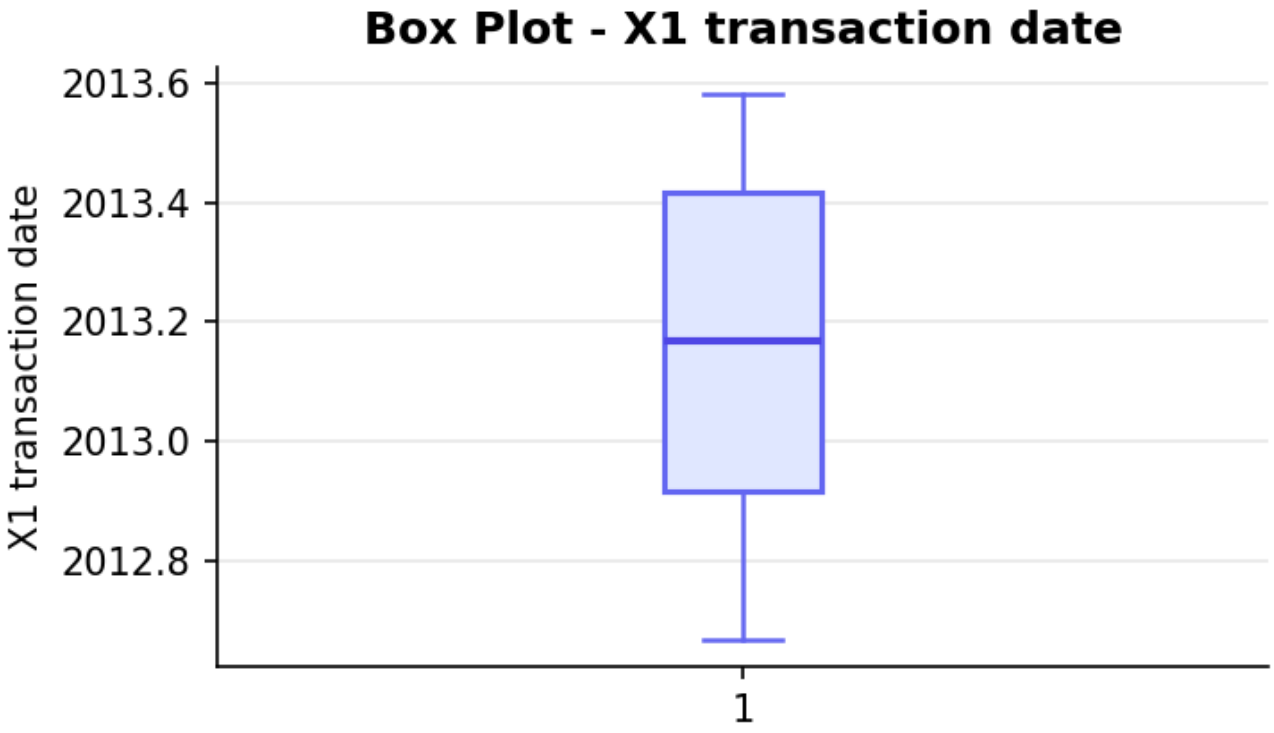
X4 number of convenience stores	0	0.0%	1.00	6.00	5.00	-6.50	13.50
X5 latitude	8	1.9%	24.96	24.98	0.01	24.94	25.00
X6 longitude	35	8.5%	121.53	121.54	0.02	121.51	121.57
Y house price of unit area	3	0.7%	27.70	46.60	18.90	-0.65	74.95

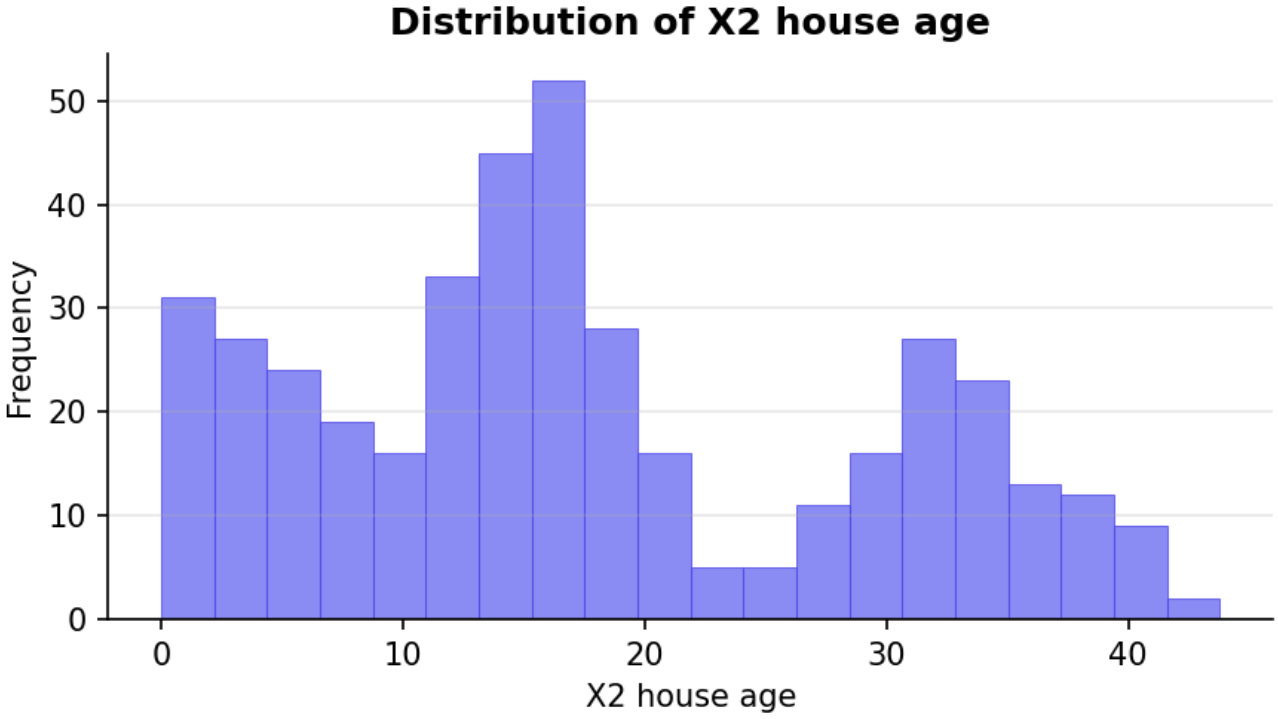
Distribution Analysis

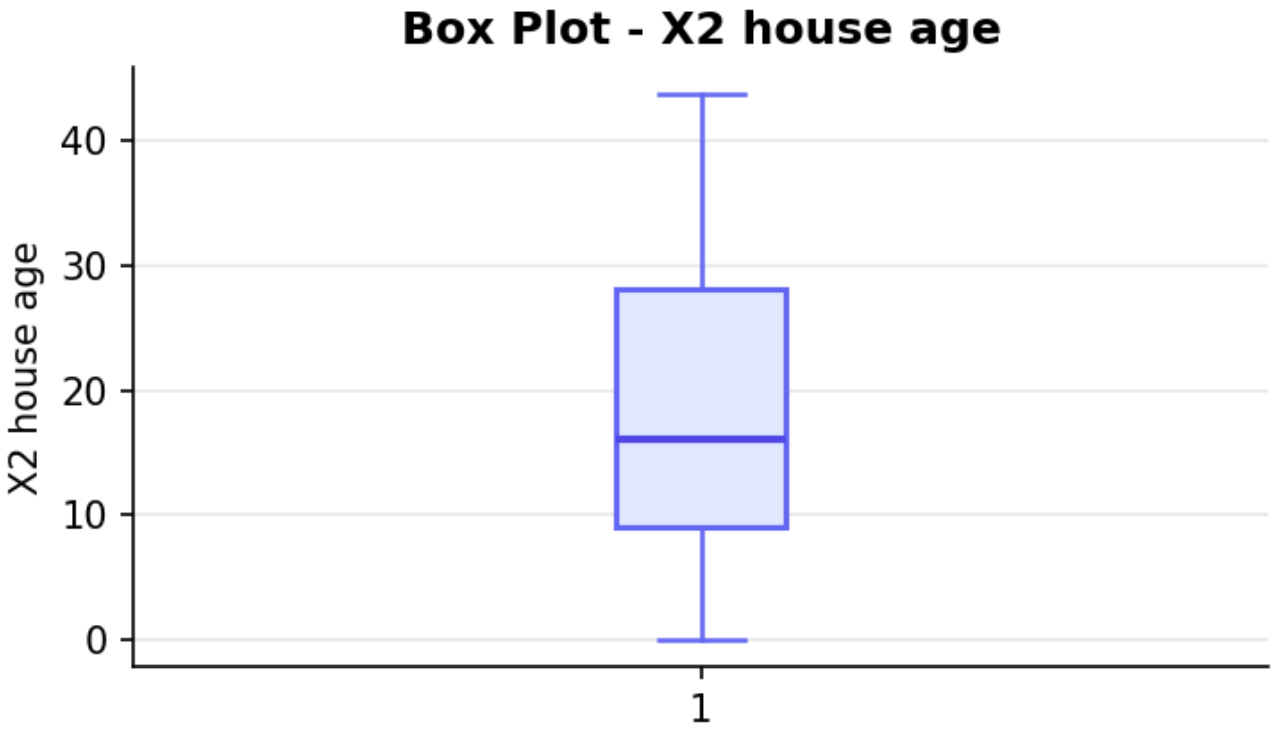


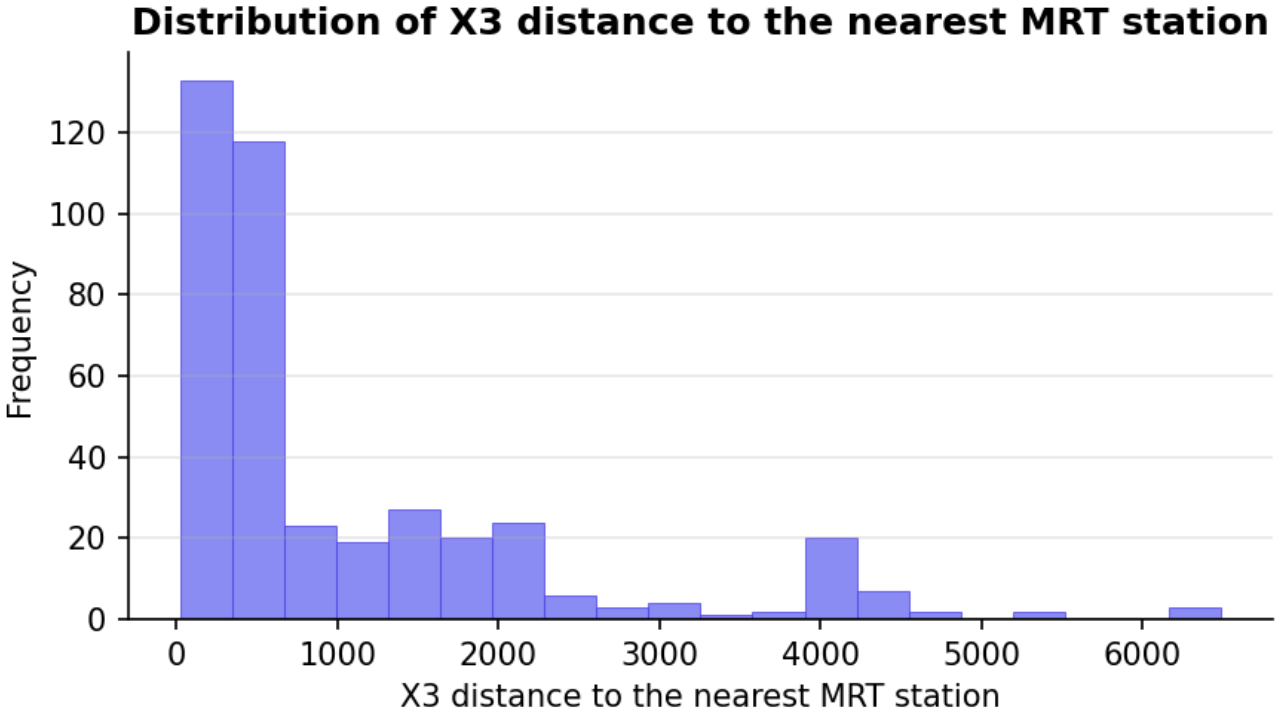

Box Plot - No
No
400
300
200
100
0
1

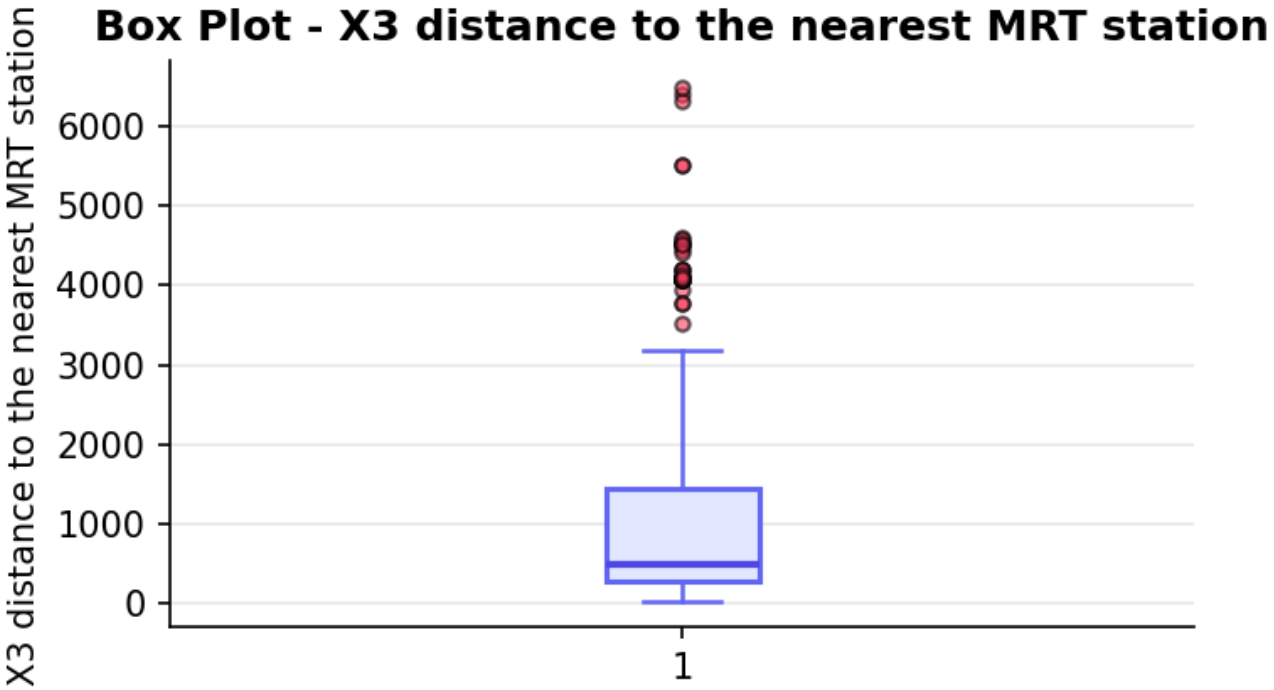

Distribution of X1 transaction date
Frequency
60
50
40
30
20
10
0
2012.8
2013.0
2013.2
2013.4
2013.6
X1 transaction date


Box Plot - X1 transaction date
X1 transaction date
2013.6
2013.4
2013.2
2013.0
2012.8
1

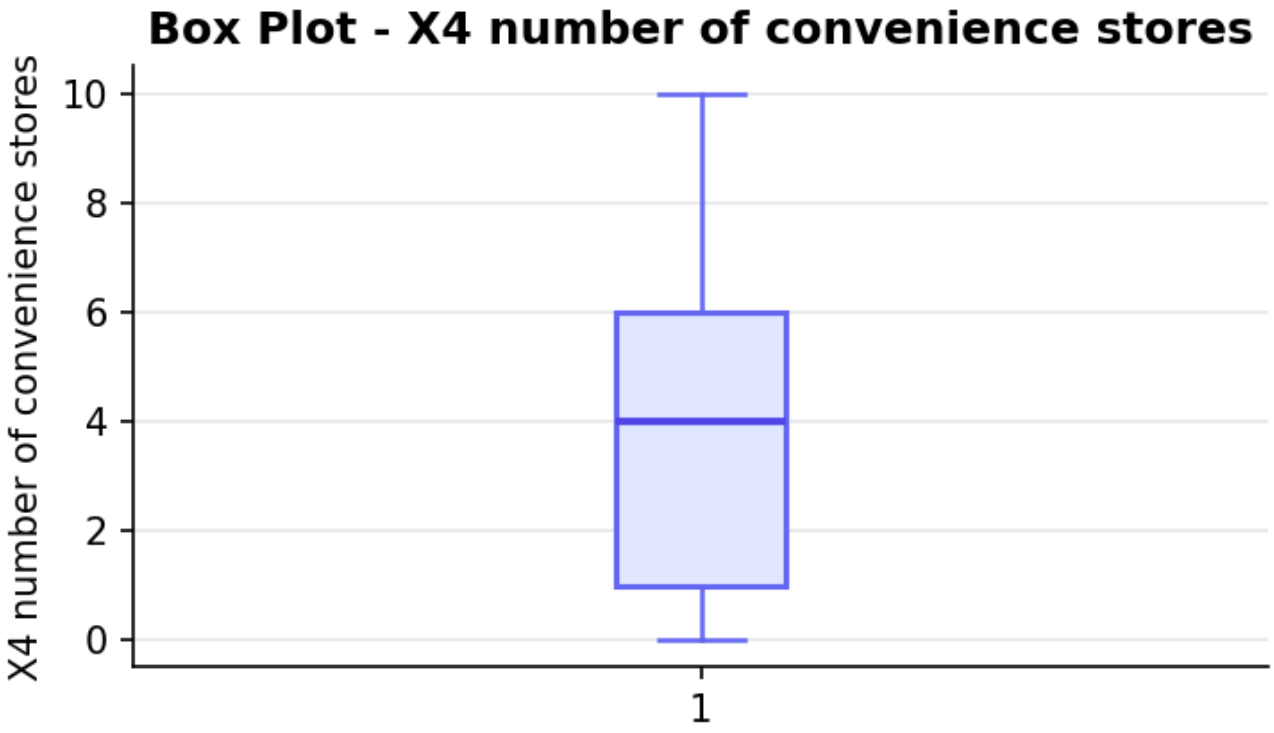

Distribution of X2 house age
Frequency
50
40
30
20
10
0
0
10
20
30
40
X2 house age

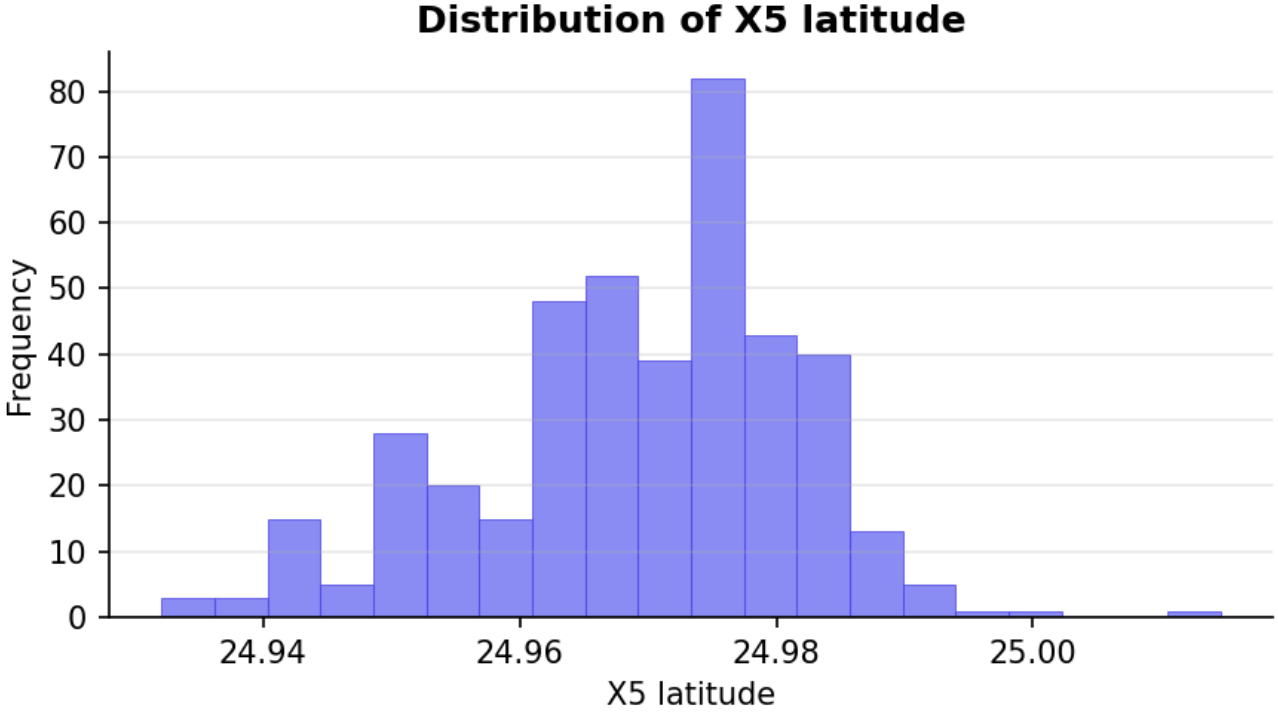

Box Plot - X2 house age
X2 house age
40
30
20
10
0
1

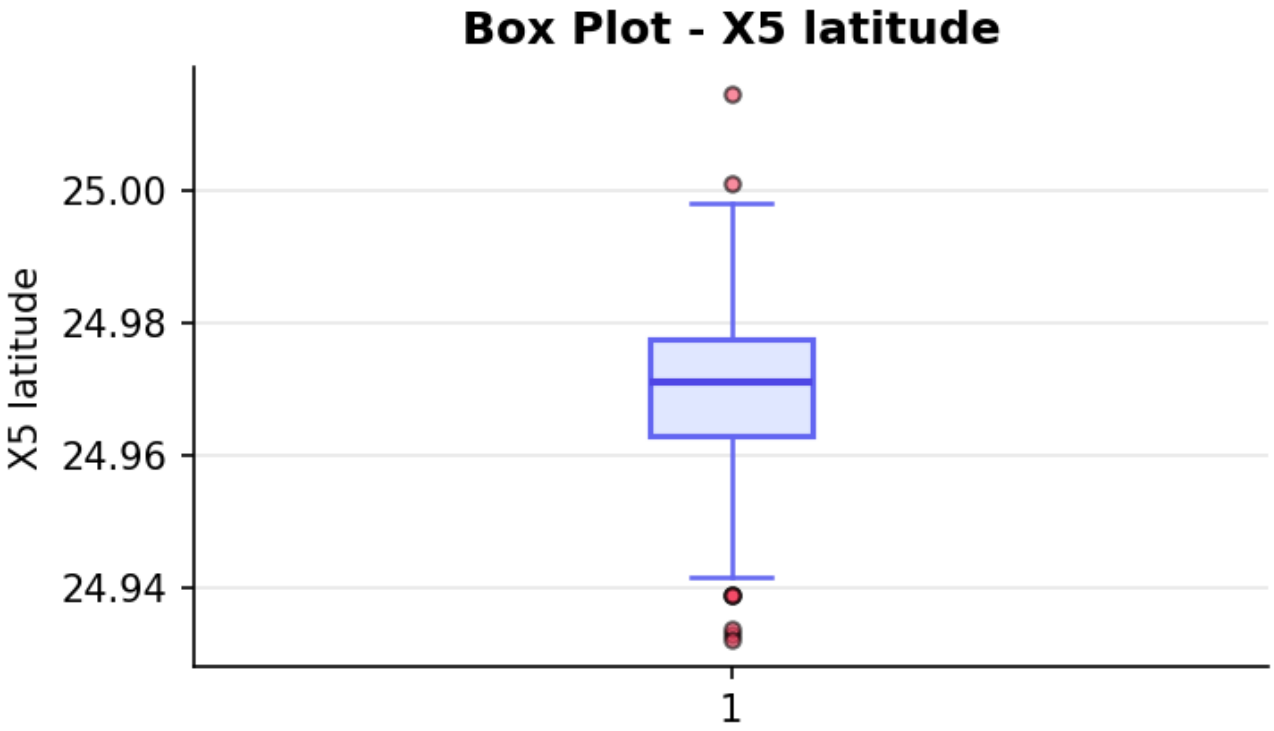

Distribution of X3 distance to the nearest MRT station
Frequency
120
100
80
60
40
20
0
0
1000
2000
3000
4000
5000
6000
X3 distance to the nearest MRT station

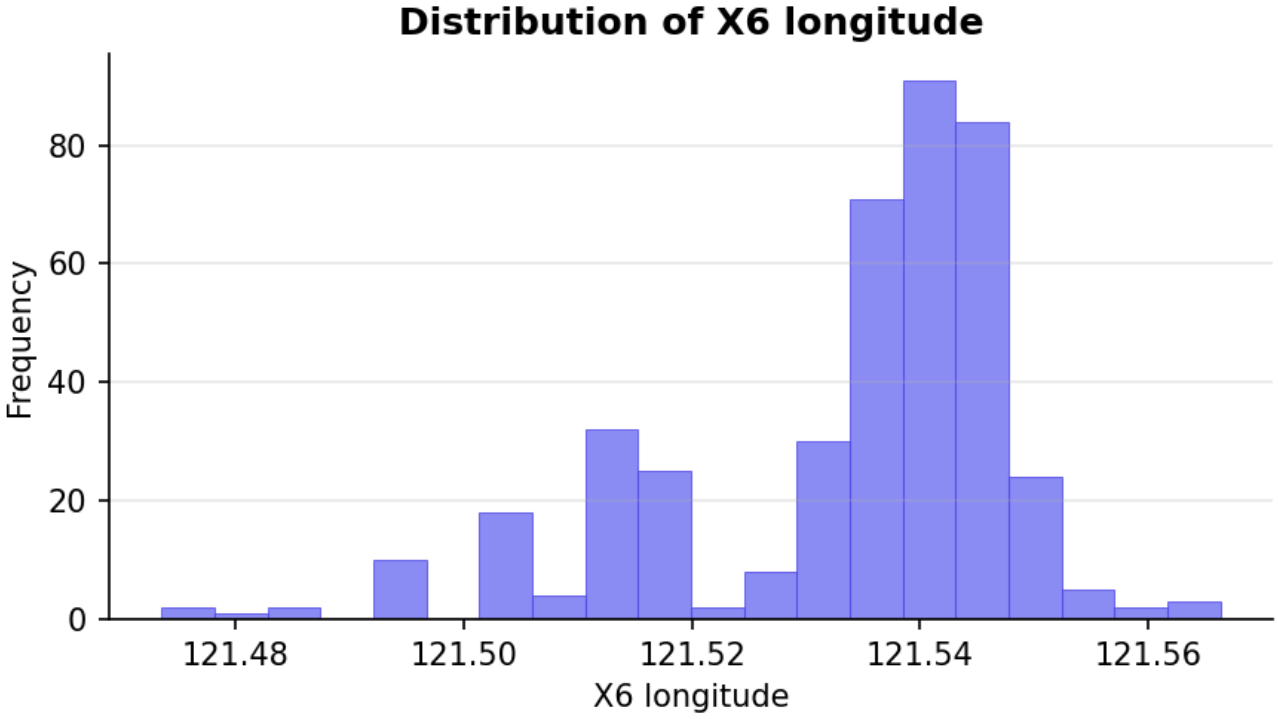

Box Plot - X3 distance to the nearest MRT station
X3 distance to the nearest MRT station
6000
5000
4000
3000
2000
1000
0
1

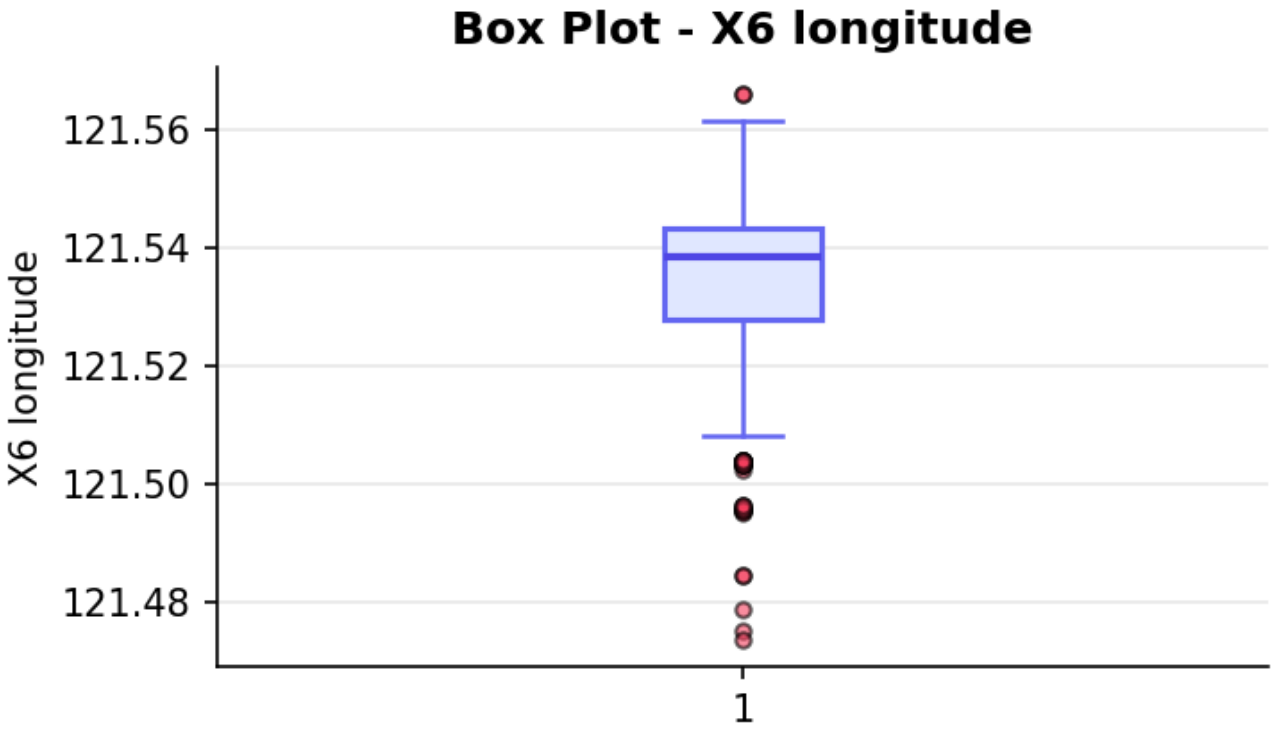

Distribution of X4 number of convenience stores
Frequency
70
60
50
40
30
20
10
0
0
2
4
6
8
10
X4 number of convenience stores

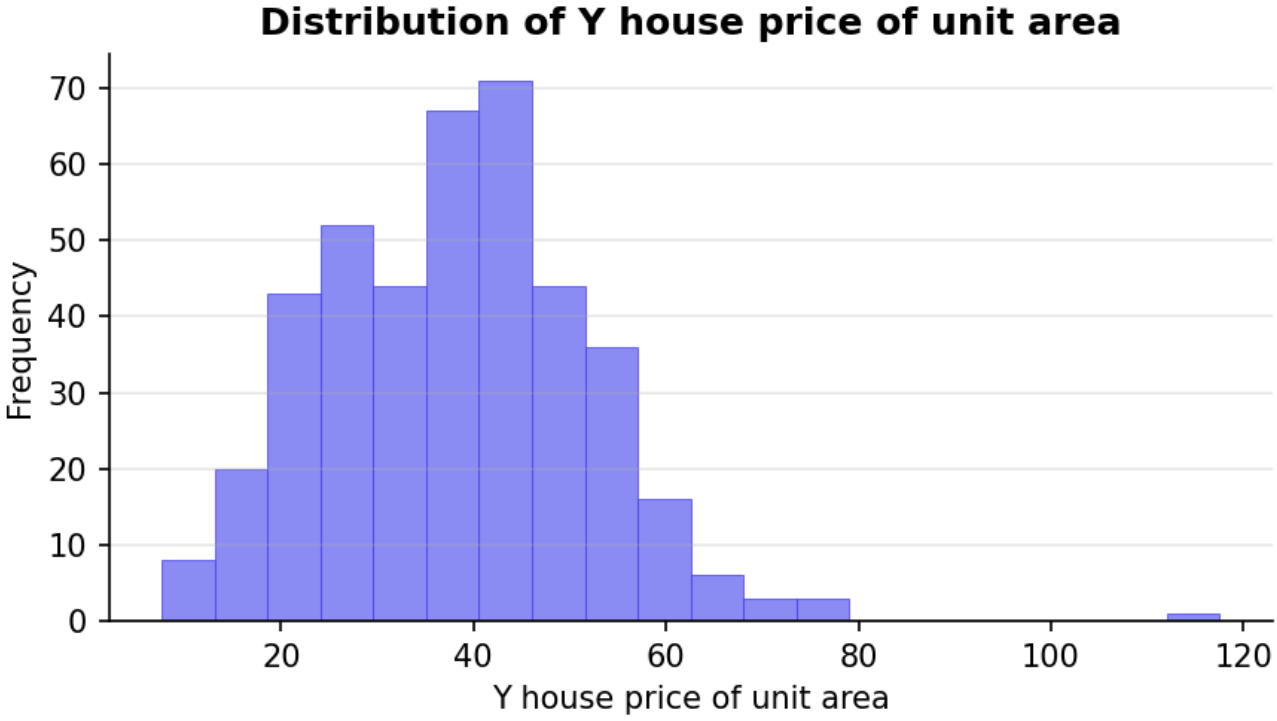

Box Plot - X4 number of convenience stores
X4 number of convenience stores
10
8
6
4
2
0
1

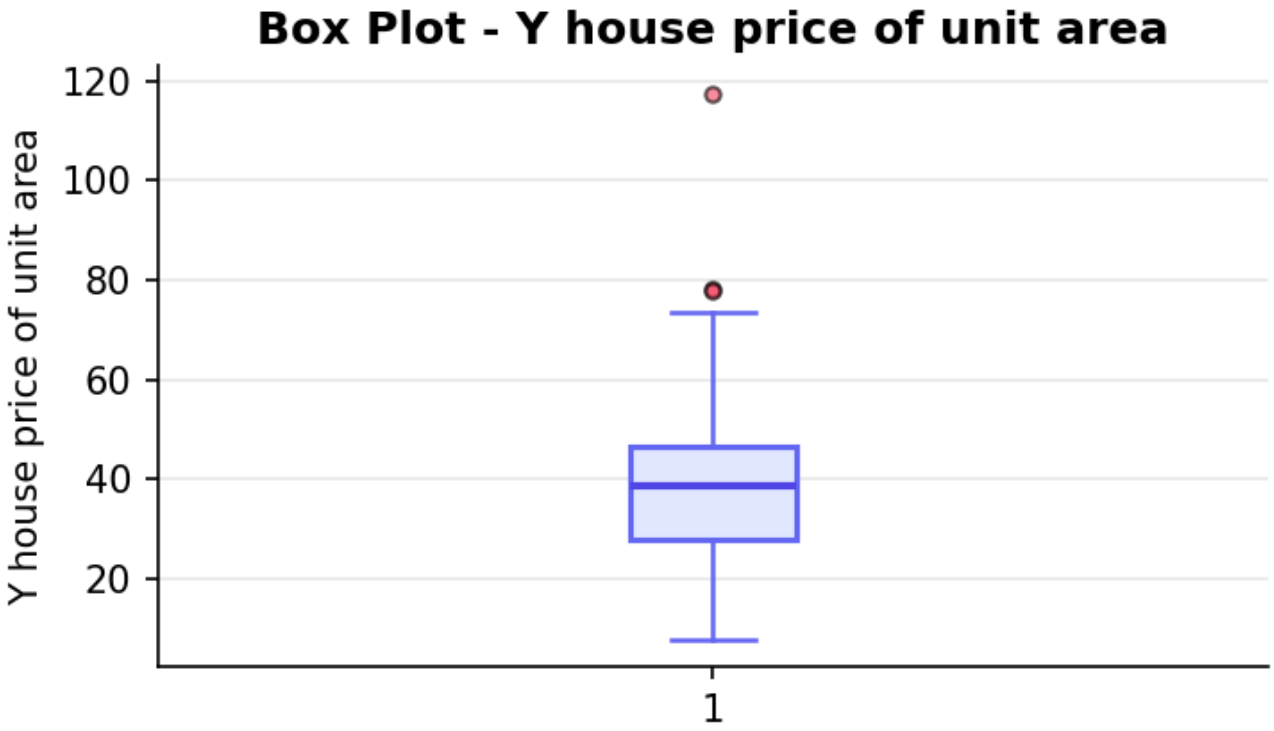

Distribution of X5 latitude
Frequency
80
70
60
50
40
30
20
10
0
24.94
24.96
24.98
25.00
X5 latitude


Box Plot - X5 latitude
X5 latitude
25.00
24.98
24.96
24.94
1


Distribution of X6 longitude
Frequency
80
60
40
20
0
121.48
121.50
121.52
121.54
121.56
X6 longitude


Box Plot - X6 longitude
X6 longitude
121.56
121.54
121.52
121.50
121.48
1


Distribution of Y house price of unit area
Frequency
70
60
50
40
30
20
10
0
20
40
60
80
100
120
Y house price of unit area


Box Plot - Y house price of unit area
Y house price of unit area
120
100
80
60
40
20
1